

Suppose, we have to arrange the fractions  $\frac{3}{5}$ ,  $\frac{6}{7}$  and  $\frac{7}{9}$  in descending order.

Now,  $\frac{3}{5} = 0.6$ ,  $\frac{6}{7} = 0.857$ ,  $\frac{7}{9} = 0.777\dots$

Since  $0.857 > 0.777\dots > 0.6$ , so  $\frac{6}{7} > \frac{7}{9} > \frac{3}{5}$ .

**VI. Recurring Decimal:** If in a decimal fraction, a figure or a set of figures is repeated continuously, then such a number is called a *recurring decimal*.

In a recurring decimal, if a single figure is repeated, then it is expressed by putting a dot on it. If a set of figures is repeated, it is expressed by putting a bar on the set.

Thus,  $\frac{1}{3} = 0.333\dots = 0.\dot{3}$ ;  $\frac{22}{7} = 3.142857142857\dots = 3.\overline{142857}$ .

**Pure Recurring Decimal:** A decimal fraction in which all the figures after the decimal point are repeated, is called a pure recurring decimal.

**Converting a Pure Recurring Decimal into Vulgar Fraction:** Write the repeated figures only once in the numerator and take as many nines in the denominator as is the number of repeating figures.

Thus,  $0.\dot{5} = \frac{5}{9}$ ;  $0.\overline{53} = \frac{53}{99}$ ;  $0.\overline{067} = \frac{67}{999}$ ; etc.

**Mixed Recurring Decimal:** A decimal fraction in which some figures do not repeat and some of them are repeated, is called a mixed recurring decimal.

e.g.,  $0.17333\dots = 0.17\overline{3}$ .

**Converting a Mixed Recurring Decimal into Vulgar Fraction:** In the numerator, take the difference between the number formed by all the digits after decimal point (taking repeated digits only once) and that formed by the digits which are not repeated. In the denominator, take the number formed by as many nines as there are repeating digits followed by as many zeros as is the number of non-repeating digits.

Thus,  $0.1\dot{6} = \frac{16-1}{90} = \frac{15}{90} = \frac{1}{6}$ ;  $0.22\overline{73} = \frac{2273-22}{9900} = \frac{2251}{9900}$ .

#### VII. Some Basic Formulae:

- $(a + b)(a - b) = (a^2 - b^2)$
- $(a + b)^2 = (a^2 + b^2 + 2ab)$
- $(a - b)^2 = (a^2 + b^2 - 2ab)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
- $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
- $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
- When  $a + b + c = 0$ , then  $a^3 + b^3 + c^3 = 3abc$ .

### SOLVED EXAMPLES

**Ex. 1.** Convert the following into vulgar fractions:

(i) 0.75                      (ii) 3.004                      (iii) .0056

**Sol.** (i)  $0.75 = \frac{75}{100} = \frac{3}{4}$ . (ii)  $3.004 = \frac{3004}{1000} = \frac{751}{250}$ . (iii)  $.0056 = \frac{56}{10000} = \frac{7}{1250}$ .

**Ex. 2.** Arrange the fractions  $\frac{5}{8}$ ,  $\frac{7}{12}$ ,  $\frac{13}{16}$ ,  $\frac{16}{29}$  and  $\frac{3}{4}$  in ascending order of magnitude.

**Sol.** Converting each of the given fractions into decimal form, we get:

$\frac{5}{8} = 0.625$ ,  $\frac{7}{12} = 0.5833$ ,  $\frac{13}{16} = 0.8125$ ,  $\frac{16}{29} = 0.5517$  and  $\frac{3}{4} = 0.75$ .

Now,  $0.5517 < 0.5833 < 0.625 < 0.75 < 0.8125$ .

$\therefore \frac{16}{29} < \frac{7}{12} < \frac{5}{8} < \frac{3}{4} < \frac{13}{16}$ .